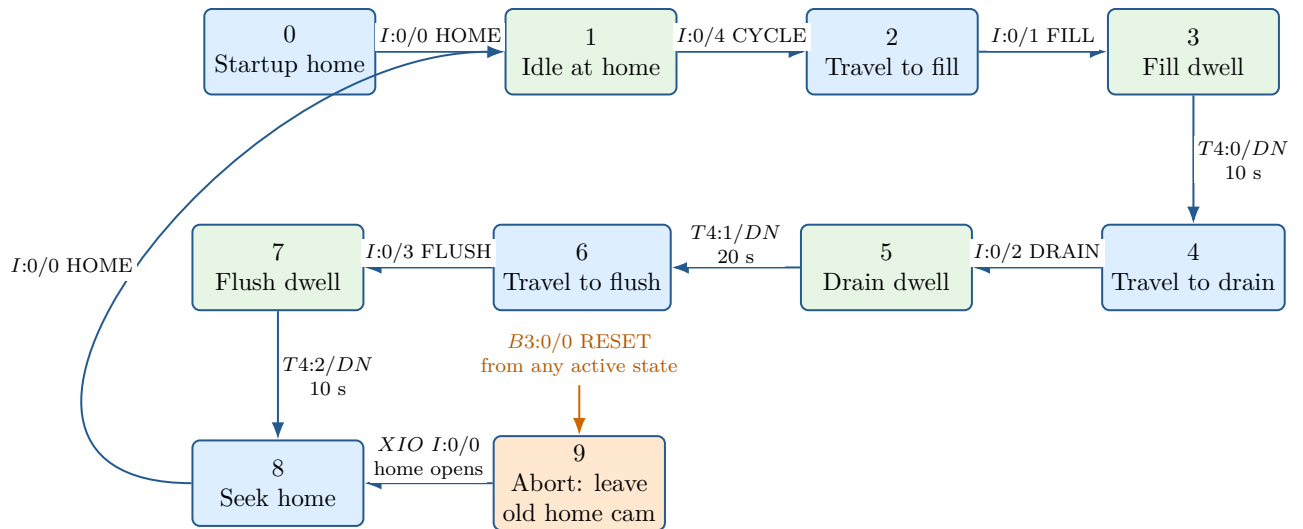


Project 4: Rotary Valve State Control

RSLogix 500 state diagram, timers, and output logic

State diagram



travelling: servo ON
 stopped/dwelling: servo OFF
 reset recovery

State and output table

Step	Meaning	Servo O:0/0	Exit condition / next step
0	Startup homing	ON	HOME I:0/0 → 1
1	Idle at home	OFF	CYCLE I:0/4 → 2
2	Travel to fill	ON	FILL I:0/1 → 3
3	Fill for 10 s	OFF	T4:0/DN → 4
4	Travel to drain	ON	DRAIN I:0/2 → 5
5	Drain for 20 s	OFF	T4:1/DN → 6
6	Travel to flush	ON	FLUSH I:0/3 → 7
7	Flush for 10 s	OFF	T4:2/DN → 8
8	Return/search for home	ON	HOME I:0/0 → 1
9	Abort; clear old home indication	ON	HOME opens (XIO I:0/0) → 8

RSLogix timer and output implementation

Timers

Enable condition	Timer	Time base	Preset
$N7:1 = 3$	$T4:0$ Fill	0.01 s	1000 (10 s)
$N7:1 = 5$	$T4:1$ Drain	0.01 s	2000 (20 s)
$N7:1 = 7$	$T4:2$ Flush	0.01 s	1000 (10 s)

Each TON resets automatically when its state changes and its enabling rung becomes false.

Servo output equation

$$O:0/0 = (N7:1 = 0) \vee (N7:1 = 2) \vee (N7:1 = 4) \vee (N7:1 = 6) \vee (N7:1 = 8) \vee (N7:1 = 9)$$

Implement the six comparisons as parallel branches driving one OTE $O:0/0$.

Reset/abort logic

Place the reset rung after the normal transition rungs so that it wins the scan:

```
XIC B3:0/0    --  MOV 9 N7:1
```

Then wait until the existing HOME cam indication opens:

```
EQU N7:1 9    XIO I:0/0    --  MOV 8 N7:1
```

Finally, accept a new HOME indication:

```
EQU N7:1 8    XIC I:0/0    --  MOV 1 N7:1
```

State 9 is required because the reset test is initiated while the original HOME microswitch may still be closed. Without state 9, the program could immediately conclude that the valve had returned home.

Position indication in $N7:0$

Execute these rungs from top to bottom:

```
MOV 4 N7:0
XIC I:0/0 -- MOV 0 N7:0
XIC I:0/1 -- MOV 1 N7:0
XIC I:0/2 -- MOV 2 N7:0
XIC I:0/3 -- MOV 3 N7:0
```

With all position switches off, the default value remains 4 (TRAVELLING). Only one position microswitch should normally be closed at a time.